

Tutorial

CNN Activation Map Calculation

Course:

Deep Learning

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Complete CNN example done in two clear phases:

1. Normal forward convolution (input → activation map)
2. Reverse calculation (activation map → original input size)

Phase 1: Normal CNN Example (Forward Calculation)

We do 1D reasoning first and it applies exactly the same to each dimension in 2D.

Given Input

- Input size: $n = 7$

Think of a 7×7 image (we calculate one dimension).

Convolution Parameters

- Filter (kernel) size: $f = 3$
- Padding: $p = 1$
- Stride: $s = 2$

Step 1: Write the Forward Formula

$$o = \left\lfloor \frac{n - f + 2p}{s} \right\rfloor + 1$$

Step 2: Substitute Values

$$o = \left\lfloor \frac{7 - 3 + 2(1)}{2} \right\rfloor + 1$$

$$o = \left\lfloor \frac{6}{2} \right\rfloor + 1$$

$$o = 3 + 1 = \boxed{4}$$

Result (Forward Pass)

Layer	Size
Input	7×7
Filter	3×3
Padding	1
Stride	2
Output (activation map)	4×4

Step 3: Intuition (Why 4?)

- Padding makes input: $7 + 2 = 9$
- Kernel slides with step size = 2
- Valid kernel positions: 0, 2, 4, 6 $\rightarrow$ **4 positions**

Phase 2: Reverse Calculation (Recover Input Size)

Now we **reverse** the calculation.

This is used in:

- Transposed convolution
- Decoder networks (U-Net, autoencoders)
- Exam questions

Given (from activation map)

- Output size: $o = 4$
- Filter size: $f = 3$
- Padding: $p = 1$
- Stride: $s = 2$

Step 1: Reverse Formula

$$n = (o - 1)s - 2p + f$$

Step 2: Substitute Values

$$n = (4 - 1) \times 2 - 2(1) + 3$$

$$n = 6 - 2 + 3$$

$$n = \boxed{7}$$

Result (Reverse Pass)

Stage	Size
Activation map	4×4
Recovered input	7×7 

 Exactly matches the original input size.

Why Reverse Formula Works (Simple Logic)

- $(o - 1) \times s \rightarrow$ total movement of kernel
- $- 2p \rightarrow$ remove padding
- $+ f \rightarrow$ add kernel size back

That's why **transposed convolution** uses the same math.

Final Summary (One Look)**Forward CNN**

$$o = \left\lfloor \frac{n - f + 2p}{s} \right\rfloor + 1$$

Reverse CNN

$$n = (o - 1)s - 2p + f$$

Verified Example

Input : 7×7

Filter : 3×3

Stride : 2

Padding: 1

Output : 4×4

Reverse: 7×7